Domain Understanding

PlaySafe

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1.Introduction

In professional football, injuries are a common problem that can affect both players and teams. When a player gets injured, they may be unable to train or play matches for a certain period. This can affect the team's performance, the coach's plans and the club's finances. For this reason, understanding which factors are related to injuries is important for professional football clubs.

Nowadays, data can help clubs make better decisions about player health and physical preparation. By analysing information from previous seasons, it may be possible to identify patterns that are related to future injuries. Machine learning can be useful for this because it allows us to analyse different variables and investigate relationships that might not be easy to identify manually.

To develop this model, I need to understand which factors are related to football injuries, what previous research has found and what information is necessary to make a prediction. This is the main purpose of this report.

2.Research Question

The main research question for this project is:

To what extent can historical player characteristics, playing time and previous injury information be used to predict whether a football player will suffer at least one injury during the following season?

To answer this question, I will investigate several aspects of football injuries. First, I will look at the factors that previous studies have associated with injuries in professional football. I will also investigate the relationship between previous injuries, playing time and player characteristics.

Another important aspect is how previous studies have used data to analyse or predict injuries. By comparing their methods and results, I can better understand which variables could be useful for my project and what limitations I need to consider.

Finally, I will use the findings to decide which information is most relevant for the prediction model will help me at the moment of implementing it.

3.Research methods

For this domain understanding, I used literature research to learn more about football injuries and injury prediction.
I selected five academic articles that investigate different aspects of injuries in professional football.

I focused on these topics because they are closely related to my project. My dataset contains information about LaLiga players, their seasons, playing time and injuries, so I wanted to understand how these variables have been investigated in previous research.

The purpose of this research is not to discover new causes of football injuries. Instead, I want to improve my understanding of the domain so I can make better decisions when designing my dataset and developing the prediction model

4.Research findings

4.1 Injuries in Professional Football

Ekstrand et al. investigated injury incidence and injury patterns in professional European football. The researchers followed professional football teams and collected information about injuries and player exposure during training sessions and matches.

The study found that injuries occurred more frequently during matches than during training when comparing the number of injuries per hour of exposure. It also showed that reinjuries resulted in longer absences than other injuries.

This research helped me understand that injuries are an important issue in professional football and that playing exposure is relevant when studying injury occurrence.

For PlaySafe, this means that information about previous injuries and playing time could be useful to investigate.

4.2 Previous Injuries as a Risk Factor

McCall et al. reviewed scientific evidence about injury risk factors, screening tests and prevention strategies in professional football.

One of the findings was that previous injury history had supporting evidence as an injury risk factor.

I found this relevant because my project uses historical player information. If previous injuries are associated with future injury risk, injury history could be an important variable to investigate.

At the same time, this does not mean that a player who has been injured before will necessarily suffer another injury. It only suggests that previous injury information may help explain differences in injury risk between players.

For PlaySafe, I want to investigate whether variables such as previous injuries and days missed because of injuries provide useful information for predicting injuries in the following season.

4.3 Playing Exposure and Workload

Jiang et al. conducted a systematic review investigating the relationship between workload and injury risk in professional male football players.

The researchers examined studies that used different workload measurements. They found that short-term fixture congestion may increase match injury incidence, but the relationship between other workload measurements and injuries was not always clear.

This helped me understand that workload is a complex topic. Playing more matches does not automatically mean that a player will get injured, and different workload measurements may be related to injuries in different ways.

For my project, I have information about playing time, such as matches played and minutes played. Although these variables are not the same as detailed physical workload measurements, they could provide some information about a player's previous playing exposure.

4.4 Machine Learning for Injury Prediction

Rossi et al. investigated how machine learning could be used to forecast injuries in professional football.

The researchers used GPS training data and machine learning to investigate whether information about players' training workloads could help predict injuries.

Their study showed that machine learning can be applied to football injury forecasting using historical training information.

I found this article particularly relevant because it is closely related to the objective of my project idea. It demonstrates how player data can be used to investigate injury risk instead of relying only on traditional methods.

However, there is an important difference between this study and my project. This study used detailed GPS training measurements, while my dataset mainly contains player characteristics, playing time and injury-related information.
This means I cannot directly apply the same approach or expect the same results.

4.5 Challenges of Predicting Football Injuries

Majumdar et al. reviewed how machine learning has been used to understand and predict injuries in football.

The paper discusses different approaches to injury prediction and the challenges involved in using training and competition load data.

One important point is that predicting injuries is difficult because injuries are influenced by several factors, and the available data may not capture all of them.

The article also helped me understand that machine learning results need to be interpreted carefully. A model may identify patterns in historical data, but this does not necessarily mean that it can predict future injuries accurately.

For PlaySafe, this means that I should not expect the model to predict every injury correctly. Instead, my goal is to investigate whether historical player information can provide useful estimates of injury probability.
It also showed me why data quality and appropriate model evaluation are important.

5. Comparison of the Research Findings

One similarity between the studies is that they all show how important it is to understand the information used when investigating football injuries.

The studies also suggest that previous injuries and playing exposure are relevant areas to investigate. However, they do not all measure these factors in the same way.

For example, Ekstrand et al. (2011) analyse injury incidence in relation to player exposure, while Jiang et al. (2022) focus more specifically on workload measurements and injury risk.

Another difference is the type of data used. Rossi et al. (2018) use detailed GPS training data, while the other studies examine injury records, risk factors or findings from previous research.

This is important for my project because I do not have access to all the variables used in these studies.

Overall, the research helped me understand that previous injury history and playing exposure are reasonable starting points for my prediction model.

6. Conclusion

Through this domain understanding, I learned more about the factors related to injuries in professional football and how previous research has investigated injury prediction.
The five articles helped me understand that injuries are a complex problem and that several factors may be associated with injury risk.

Previous injury history appears to be an important factor to investigate, while playing exposure and workload may also provide useful information. The machine learning studies show that historical data can be used to investigate injury prediction, although the results depend on the quality and type of information available.

For PlaySafe, I decided to focus on predicting whether a LaLiga player will suffer at least one injury during the following season.

The research helped me identify previous injuries, playing exposure and player characteristics as potential predictors. However, I still need to analyse my dataset to determine which variables are actually useful.

I also learned that defining the target variable correctly is especially important. Missing injury information needs to be handled carefully, and the model must use only historical information to avoid data leakage.

Overall, this research gives me a better understanding of the problem I am trying to solve and provides a starting point for the data design and modelling phases of PlaySafe.

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